



Acoustic and thermophysical aspects of AeZrH₃ Perovskites for efficient hydrogen storage via d-band center: DFT+U study with van der Waals interactions

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ABSTRACT

Perovskite materials are a promising candidate for solid-state hydrogen storage applications. Based on a first-principles study, the newly formed AeZrH₃ (Ae = Li/Na/K/Rb/Cs) perovskite hydrides were investigated using van der Waals interactions within the DFT + D approach. This work systematically studies the acoustic and thermophysical aspects of the hydrogen storage characteristics of AeZrH₃ using the density functional theory (DFT) method. The tolerance factor suggests these materials are in the range of stable cubic structures, and AIMD simulation results further confirm their dynamical stability under finite-temperature conditions. With and without spin-orbit coupling (SOC), electronic results show that all hydrides exhibit metallic behavior with a high contribution of Zr d-state. DFT + U analysis indicated KZrH₃, RbZrH₃, and CsZrH₃ have small band gap energies (0.77, 0.82, and 0.99 eV), and the rest (LiZrH₃, NaZrH₃) are metallic compounds, confirmed by electronic properties. Mulliken charge analysis shows the charge transfer characteristics of these compounds, demonstrating that charges move from the Ae and Zr to the H atoms, which makes covalent bonding. AeZrH₃ hydrides exhibited the d-band center, showing a significant upshift for hydrogen uptake and covalent bonding. The elastic properties revealed that NaZrH₃ is the hardest material among them and has a high Debye temperature and thermal conductivity. This study shows the novel AeZrH₃ compounds, providing valuable insights into hydrogen storage applications.

1. Introduction

The two biggest drivers in scientific research today are the need for a pollution-free environment and the energy crisis, which highlights the need to improve the human condition and the planet's growing population [1,2]. Hydrogen is now being viewed as a viable fuel alternative to the power generation industry due to its high energy density, cleanliness, light weight, zero pollutants, and potential for generation with renewable sources [3]. Hydrogen is also unlimited as an energy source compared to fossil fuels, which are finite for stationary and mobile uses.

The biggest hurdle to putting hydrogen-powered vehicles on the market is developing safe but effective storage methods. The current energy crisis and environmental degradation are going to change the case for the energy crisis, the increasing energy consumption, the depletion of fossil fuels, and the degradation of the environment [4]. Technological advancements have led to increased energy consumption in modern societies, further straining the limited availability of fossil fuels. Currently, fossil fuels remain the primary energy source, but their finite nature necessitates the exploration of alternatives. To address these challenges, the US Department of Energy (DOE) has established

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standards for hydrogen storage techniques. These standards include specific requirements for gravimetric and volumetric storage capacities, temperature ranges (233 to 333 K), and pressures (up to 10 MPa) to be achieved by 2025 [5]. Given the gravimetric and volumetric minima of 5.5 wt% and 40 gH₂/L, respectively, solid-state energy storage indicates energy stored is higher than opposed of liquids or gases [6,7]. Solid-state is advantageous because the solid nature entails higher energy densities, safer operation, and longer cycle life compared to conventional liquid electrolyte-based systems [8]. For example, solid-state batteries use solid electrolytes and electrodes to store and release energy [9]. That said, solid-state batteries can revolutionize energy storage from consumer electronics to electric vehicles to grid-scale energy storage. Other solid-state methods of energy storage include supercapacitors, which store energy through the physical separation of charges at interfaces between solid materials and electrolytes [6,10]. Researchers are investigating solid-state energy storage materials to improve performance metrics, including energy density, charging rates, and cycle life, to facilitate the adoption of renewable energy and the electrification of transportation.

Recently, perovskite hydrides have received a lot of attention due to their potential applications in the field of science and technology, like energy storage, sensors, supercapacitors, fuel cells, and optoelectronics [11–14]. NaMgH₃ hydrides were experimentally studied and found that during the decomposition reaction, the enthalpy and entropy are changed [15]. Martínez-Coronado et al. [16] experimentally synthesized cubic Na_{1-x}Li_xMgH₃ hydride and obtained a desorption temperature in the range of 413–435 °C. The MMgH₃ hydrides were synthesized by the ball milling method; they found that the RbMgH₃ is stable among other compounds [17]. Xu, N. et al. [18] studied the XTiH₃ perovskite-type hydrides for hydrogen storage applications by using the DFT method. They have obtained the maximum gravimetric capacity, 3.36, wt%, respectively. XAlH₃ hydrides were studied for hydrogen storage applications [19]. They found that NaAlH₃ has ductility and KAlH₃ has brittle materials. The calculated gravimetric capacities for NaAlH₃ and KAlH₃ are 5.40 and 4.19 wt%. The new perovskite hydrides XPtH₃ were investigated by using the DFT method, and determined that the hydrogen storage densities are 1.45, 1.35, 1.26, and 1.06 wt%, respectively [20]. Moreover, different hydrides have been systematically studied for hydrogen storage applications like as LiXH₃ [21], LiXH₃ (X = Ca, Sr, Ba) [22], MgXH₃ (X = Al, Sc, Zr) [23], MgXH₃ (X = Cr, Mn, Fe) [24], XFeH₃ (X = Ca, Sr, Ba) [25], XPtH₃ (X = Li, Na, K, Rb) [20], XCuH₃ (X = Rb, K) [26], XSnH₃ (X = Rb, Cs, Fr) [27], CaXH₃ (X = Cu, Zn) [28], and XCrH₃ (X = Na, K) [29], etc. Beyond perovskite hydrides, several complex hydride families such as A₃BH₆, A₂BH₆, and AFe₃H₈ systems have demonstrated diverse structural frameworks and hydrogen storage capacities ranging from moderate to high gravimetric percentages [30, 31]. Besides this, we were interested in investigating zirconium-based hydrides for hydrogen storage applications and transportation. Moreover, they absorb a significant amount of hydrogen, forming zirconium hydrides (ZrH_x), have an excellent capacity to absorb and desorb hydrogen, good kinetics reversibly, and are chemically stable, as well as the d-band center has a significant role in hydrogen uptake and boosting the storage capacity [32].

These are novel perovskite hydrides, various Zr-based has been investigated for hydrogen [33,34] but up to date, theoretical (DFT) and experimental studies on the acoustic, thermophysical aspects, d-band center interaction, Spin-orbit coupling (SOC), and DFT + U approaches of AeZrH₃ (Ae = Li/Na/K/Rb/Cs) for hydrogen storage applications have not been carried out. Therefore, the DFT study was employed for the aforementioned characteristics of AeZrH₃ lightweight perovskite hydrides. The findings were meticulously analyzed and discussed to provide a deeper understanding of the physical characteristics of these materials as promising candidates for solid-state hydrogen storage. This work is structured as follows: Section 2: details the computational methods and procedures used in the study. Section 3: presents and discusses the results related to physical properties. Section 4: d-band center

role and hydrogen storage application, and finally, Section 5: summarizes our computational findings.

2. Computational Methodology

The computational results were conducted on the CASTEP code based on DFT studies [35]. The van der Waals dispersion corrections were implemented by the DFT-D method of the Grimme type and were combined with the GGA-PBE exchange-correlation functional [36]. The core-electron interactions were described by a projector-augmented wave technique [37]. The wave functions were expanded by a plane-wave basis set with a cutoff energy of 680 eV. The Brillouin zone integrations were generated from a Monkhorst-Pack k-point mesh of 8 × 8 × 8 (the spacing of the distance was 0.031 Å⁻¹). After the convergence, the average results of 0.5 × 10⁻⁶ eV for electronic total energy, 0.1 × 10⁻¹ eV/Å for atomic forces, and 0.01 GPa pressure were obtained for all materials at room temperature. Electronic valence shells of the individual atoms that compose the actual hydrides are Li (2s¹), Na (3s¹), K (4s¹), Rb (5s¹), Cs (6s¹), Zr (4 d² 5s²), and H (1s¹). The electronic properties were calculated using a high-dense k-point 12 × 12 × 12 for better electronic density of states. The total convergence energy is 0.5 × 10⁻⁶ eV was achieved. However, we performed the spin-orbit coupling (SOC) calculation of the electronic band structure by using the norm-conserving pseudopotential [38]. Furthermore, the DFT + U method was employed for these materials for the better visualization of valence and conduction bands. The U = 4 eV value was set for the Zr-d state, which was obtained from the literature [39]. The mechanical properties were obtained from the optimized geometry based on a stress-strain method, with a maximum strain amplitude is 0.003. The optical properties were calculated using the k-point mesh 12 × 12 × 12 with the optimized GGA-PBE structure. The optical dielectric functions, refractive index, reflectivity, and adsorption were calculated by eq- (1-5) [14].

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \quad (1)$$

$$n(\omega) = \frac{1}{\sqrt{2}} \left[(\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega))^{1/2} + \varepsilon_1(\omega) \right]^{1/2} \quad (2)$$

$$k(\omega) = \frac{1}{\sqrt{2}} \left[(\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega))^{1/2} - \varepsilon_1(\omega) \right]^{1/2} \quad (3)$$

$$R(\omega) = \frac{(n-1)^2 + k^2}{(n+1)^2 + k^2} \quad (4)$$

$$\alpha(\omega) = \sqrt{2} \omega \left[(\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega))^{1/2} - \varepsilon_1(\omega) \right]^{1/2} \quad (5)$$

Additionally, X-ray diffraction (XRD) patterns were calculated from the relaxation structure and viewed on VESTA software using the wavelength of Cu Kα 1.540 Å [3]. Furthermore, Ab initio molecular dynamics (AIMD) simulations using the VASP software were used to investigate the finite-temperature dynamical stability of all AeZrH₃ (Ae = Li, Na, K, Rb, Cs) hydrides [40,41]. A Nosé–Hoover thermostat was used to maintain the temperature at 300 K while the simulations were run in the canonical NVT ensemble. To provide sufficient sampling of thermal fluctuations, each system was developed for 5 ps (5000 fs) using a time step of 1 fs.

3. Results and discussions

3.1. Structural and materials stability properties

Fig. 1 illustrates the cubic crystal structure of AeZrH₃ perovskite hydride, with the space group number being the *Pm* $\bar{3}$ *m* (#221) [19], and the atomic coordinates of Ae, Zr, and H are 1a (0, 0, 0), 1b (1/2, 1/2, 1/2), and 1c (0, 1/2, 1/2), respectively. Figure S1 shows all the